CSA0412-OPERATING SYSTEMS

EXP 03-Producer and Consumer in C

```
#include <stdio.h> #include <pthread.h> #include <semaphore.h>

int buffer[5], in = 0, out = 0; sem_t empty, full, mutex;

void *producer(void *arg) { int i; for (i = 1; i <= 5; i++) { sem_wait(&empty);
sem_wait(&mutex);

   buffer[in] = i;
    printf("Produced: %d\n", i);
    in = (in + 1) % 5;

    sem_post(&mutex);
    sem_post(&full);
}
return NULL;


}

void *consumer(void *arg) { int i, item; for (i = 1; i <= 5; i++) { sem_wait(&full);
sem_wait(&mutex);

   item = buffer[out];
    printf("Consumed: %d\n", item);
    out = (out + 1) % 5;

    sem_post(&mutex);
    sem_post(&empty);
}
return NULL;


}

int main() { pthread_t p, c;

sem_init(&empty, 0, 5);
sem_init(&full, 0, 0);
sem_init(&mutex, 0, 1);

pthread_create(&p, NULL, producer, NULL);
pthread_create(&c, NULL, consumer, NULL);
```

```
pthread_join(p, NULL);
pthread_join(c, NULL);

sem_destroy(&empty);
sem_destroy(&full);
sem_destroy(&mutex);

return 0;


}
```

